Lesson 18: Compactness and the Baire Category Theorem

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

Contents

6 Compactness **2**

6.1 Definitions and Basic Theory 2

6.2 Riesz's Lemma and Compactness in Infinite Dimensions 3

6.3 The Arzelà–Ascoli Theorem 4

6.4 Compact Operators 5

7 The Baire Category Theorem **5**

7.1 Nowhere Dense Sets and Category 5

7.2 Statement and Proof 6

7.3 Application Preview 6

8 Exercises **7**

9 Summary of Key Results **8**

6 Compactness

Compactness is the key tool for passing from local to global properties and for establishing existence results. In infinite-dimensional spaces, compactness behaves very differently from the finite-dimensional case.

6.1 Definitions and Basic Theory

Definition 6.1 (Compact Set)**.** A subset K of a metric space (X, d) is *compact* if every open cover of K has a finite subcover: whenever $K \subseteq \bigcup_{\alpha \in I} G_\alpha$ with each G_α open, there exist finitely many $\alpha_1, \ldots, \alpha_n$ with $K \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}$.

Definition 6.2 (Sequential Compactness)**.** A subset K of a metric space is *sequentially compact* if every sequence in K has a subsequence that converges to a point in K.

Definition 6.3 (Totally Bounded)**.** A subset A of a metric space (X, d) is *totally bounded* if for every $\varepsilon > 0$, A can be covered by finitely many open balls of radius ε: there exist $x_1, \ldots, x_n \in X$ with $A \subseteq \bigcup_{i=1}^n B(x_i, \varepsilon)$.

Proposition 6.1 (Totally Bounded Implies Bounded)**.** *Every totally bounded set is bounded. The converse fails in general.*

Proof. If $A \subseteq \bigcup_{i=1}^n B(x_i, 1)$, then for any $a, b \in A$, $a \in B(x_i, 1)$ and $b \in B(x_j, 1)$ for some i, j, so $d(a, b) \leq d(a, x_i) + d(x_i, x_j) + d(x_j, b) < 1 + \max_{i,j} d(x_i, x_j) + 1$. Hence $\operatorname{diam}(A) < \infty$.

For the failure of the converse: the closed unit ball in ℓ^2 is bounded but not totally bounded (the standard basis vectors e_n satisfy $\|e_n - e_m\| = \sqrt{2}$ for $n \neq m$, so no finite collection of balls of radius $\sqrt{2}/2$ can cover them all). □

Theorem 6.2 (Characterization of Compactness in Metric Spaces)**.** *For a subset K of a metric space (X, d), the following are equivalent:*

(i) K is compact.
(ii) K is sequentially compact.
(iii) K is complete and totally bounded.

Proof. **(i) ⇒ (ii):** Let (x_n) be a sequence in K. Suppose for contradiction that (x_n) has no convergent subsequence in K. Then every point $y \in K$ has a neighborhood $B(y, r_y)$ containing only finitely many terms of (x_n) (otherwise, we could extract a subsequence converging to y). The collection $\{B(y, r_y) : y \in K\}$ covers K, so by compactness there is a finite subcover. But a finite union of sets, each containing finitely many terms, contains only finitely many terms total, contradicting the fact that (x_n) has infinitely many terms in K.

(ii) ⇒ (iii): *Completeness:* Let (x_n) be Cauchy in K. By sequential compactness, (x_n) has a subsequence converging to some $x \in K$. A Cauchy sequence with a convergent subsequence converges (Theorem **??** proof argument), so $x_n \to x \in K$.

Total boundedness: Suppose K is not totally bounded. Then there exists $\varepsilon > 0$ such that K cannot be covered by finitely many ε-balls. Pick $x_1 \in K$ arbitrarily. Having chosen $x_1, \ldots, x_n$, since $K \not\subseteq \bigcup_{i=1}^n B(x_i, \varepsilon)$, choose $x_{n+1} \in K \setminus \bigcup_{i=1}^n B(x_i, \varepsilon)$. Then $d(x_{n+1}, x_i) \geq \varepsilon$ for all $i \leq n$. The sequence (x_n) has no convergent subsequence (any two terms are at least ε apart), contradicting sequential compactness.

(iii) ⇒ (i): Suppose K is complete and totally bounded. We use the *Lebesgue covering lemma* approach.

Claim (Lebesgue number): For every open cover $\{G_\alpha\}$ of K, there exists $\delta > 0$ (the *Lebesgue number*) such that for every $x \in K$, $B(x, \delta) \subseteq G_\alpha$ for some α.

Proof of claim: Suppose not. Then for each n, there exists $x_n \in K$ with $B(x_n, 1/n) \not\subseteq G_\alpha$ for any α. By (iii) ⇒ (ii) (which we now verify: complete + totally bounded ⇒ sequentially

compact, see below), (x_n) has a subsequence $x_{n_k} \to x \in K$. Since $x \in K \subseteq \bigcup_\alpha G_\alpha$, we have $x \in G_\beta$ for some β. Since G_β is open, $B(x,r) \subseteq G_\beta$ for some $r > 0$. For large k, $d(x_{n_k}, x) < r/2$ and $1/n_k < r/2$, so $B(x_{n_k}, 1/n_k) \subseteq B(x,r) \subseteq G_\beta$, contradicting the choice of x_{n_k}.

To complete the argument, we need: complete + totally bounded implies sequentially compact. Let (x_n) be a sequence in K. Cover K by finitely many balls of radius 1; one ball contains infinitely many terms, giving a subsequence $(x_n^{(1)})$ with diam ≤ 2. Cover K by balls of radius $1/2$; one contains infinitely many terms of $(x_n^{(1)})$, giving $(x_n^{(2)})$ with diam ≤ 1. Continue: at stage k, get $(x_n^{(k)})$ contained in a ball of radius $1/k$. The diagonal subsequence $y_k = x_k^{(k)}$ is Cauchy (for $k, m \geq N$, $d(y_k, y_m) \leq 2/N$), hence converges by completeness.

Now: given the Lebesgue number $\delta > 0$, cover K by finitely many balls of radius $\delta/2$ (by total boundedness). Each ball has diameter $\leq \delta$, so is contained in some G_α. Taking the corresponding finitely many G_α's gives a finite subcover. □

Corollary 6.3 (Heine–Borel for $\mathbb{R}^n$)**.** *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. ($\Rightarrow$) Compact subsets of metric spaces are closed and bounded. (*Closed:* If $x_n \in K$ and $x_n \to x$, then by sequential compactness there is a subsequence converging in K; by uniqueness of limits, $x \in K$. *Bounded:* Compact implies totally bounded implies bounded.)

($\Leftarrow$) Let $K \subseteq \mathbb{R}^n$ be closed and bounded. Bounded in $\mathbb{R}^n$ implies totally bounded (cover K by a grid of cubes of side $\varepsilon/\sqrt{n}$). Closed subset of a complete space is complete. By Theorem 6.2, K is compact. □

6.2 Riesz's Lemma and Compactness in Infinite Dimensions

Lemma 6.4 (Riesz's Lemma)**.** *Let Y be a proper closed subspace of a normed space X (i.e., $Y \neq X$). For every $\theta \in (0,1)$, there exists $x_\theta \in X$ with $\|x_\theta\| = 1$ and $d(x_\theta, Y) = \inf_{y \in Y} \|x_\theta - y\| \geq \theta$.*

Proof. Since $Y \neq X$, there exists $z \in X \setminus Y$. Since Y is closed, $d = d(z, Y) = \inf_{y \in Y} \|z - y\| > 0$. Choose $y_0 \in Y$ with $\|z - y_0\| < d/\theta$ (possible since $d/\theta > d = \inf$). Define

$$x_\theta = \frac{z - y_0}{\|z - y_0\|}.$$

Then $\|x_\theta\| = 1$. For any $y \in Y$:

$$\|x_\theta - y\| = \frac{\|z - y_0 - \|z - y_0\| y\|}{\|z - y_0\|} = \frac{\|z - \underbrace{(y_0 + \|z - y_0\| y)}_{\in Y}\|}{\|z - y_0\|} \geq \frac{d}{\|z - y_0\|} > \frac{d}{d/\theta} = \theta.$$

Taking the infimum over $y \in Y$: $d(x_\theta, Y) \geq \theta$. □

Remark 6.1. We cannot always achieve $\theta = 1$ (i.e., exact distance 1 from Y). This is possible in reflexive spaces but not in general.

Theorem 6.5 (Riesz's Theorem: Compactness Characterizes Finite Dimension)**.** *The closed unit ball $\overline{B}(0,1) = \{x \in X : \|x\| \leq 1\}$ in a normed space X is compact if and only if X is finite-dimensional.*

Proof. ($\Leftarrow$) If $\dim X = n < \infty$, then X is isomorphic to $\mathbb{R}^n$ (Theorem **??**), and the closed unit ball is closed and bounded, hence compact by Heine–Borel (Corollary 6.3).

($\Rightarrow$) We prove the contrapositive: if X is infinite-dimensional, then $\overline{B}(0,1)$ is not compact (not sequentially compact). We construct a sequence in $\overline{B}(0,1)$ with no convergent subsequence.

Pick $x_1 \in X$ with $\|x_1\| = 1$. Let $Y_1 = \operatorname{span}\{x_1\}$, which is a proper closed subspace (finite-dimensional subspaces are closed). By Riesz's lemma with $\theta = 1/2$, there exists x_2 with $\|x_2\| = 1$ and $\|x_2 - y\| \geq 1/2$ for all $y \in Y_1$; in particular $\|x_2 - x_1\| \geq 1/2$.

Let $Y_2 = \text{span}\{x_1, x_2\}$, another proper closed subspace. By Riesz's lemma, there exists x_3 with $\|x_3\| = 1$ and $d(x_3, Y_2) \geq 1/2$. Continuing inductively, we obtain a sequence (x_n) in $\overline{B}(0,1)$ with $\|x_n - x_m\| \geq 1/2$ for all $n \neq m$.

No subsequence of (x_n) can be Cauchy (hence none can converge), so $\overline{B}(0,1)$ is not sequentially compact, hence not compact. □

Riesz's Theorem

The closed unit ball in a normed space is compact if and only if the space is finite-dimensional. This is one of the most important structural results in functional analysis: it says that in infinite-dimensional spaces, bounded sets are never compact (in the norm topology). This forces us to use weaker topologies or stronger conditions (like equicontinuity) to obtain compactness.

Compactness and Function Approximation

Riesz's theorem explains a fundamental challenge in RL: the space of all value functions on an infinite state space is infinite-dimensional, so the unit ball is not compact. This means that a bounded sequence of value function approximations need not have a convergent subsequence. In practice, this is why we restrict to finite-dimensional parameterizations (linear function approximation, neural networks) or use compactness criteria like Arzelà–Ascoli (Theorem 6.6) to guarantee that approximation sequences converge.

6.3 The Arzelà–Ascoli Theorem

Definition 6.4 (Equicontinuity)**.** A family $\mathcal{F} \subseteq C[a,b]$ is *equicontinuous* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(s) - f(t)| < \varepsilon$ for all $f \in \mathcal{F}$ and all $s, t \in [a,b]$ with $|s - t| < \delta$. The family is *uniformly bounded* if $\sup_{f \in \mathcal{F}} \|f\|_\infty < \infty$.

Theorem 6.6 (Arzelà–Ascoli)**.** *A subset $\mathcal{F} \subseteq C[a,b]$ (with the sup norm) is relatively compact (i.e., $\text{cl}(\mathcal{F})$ is compact) if and only if $\mathcal{F}$ is uniformly bounded and equicontinuous.*

Proof. ($\Leftarrow$) **Uniformly bounded + equicontinuous implies relatively compact.** By Theorem 6.2, it suffices to show that every sequence (f_n) in $\mathcal{F}$ has a uniformly convergent subsequence (the closure inherits completeness from $C[a,b]$).

Step 1: Pointwise convergence on a dense set. Let $D = \{r_1, r_2, \ldots\}$ be a countable dense subset of $[a,b]$ (e.g., the rationals in $[a,b]$). The sequence $(f_n(r_1))$ is bounded (by uniform boundedness), so by Bolzano–Weierstrass it has a convergent subsequence $(f_{n_k^{(1)}}(r_1))$. From this subsequence, extract a further subsequence convergent at r_2: $(f_{n_k^{(2)}})$. Continue for each r_j. The diagonal subsequence $g_k = f_{n_k^{(k)}}$ converges at every $r_j \in D$.

Step 2: Uniform convergence. Given $\varepsilon > 0$, choose $\delta > 0$ from equicontinuity for $\varepsilon/3$. Cover $[a,b]$ by finitely many intervals of length $< \delta$, and pick a point $r_{j_i} \in D$ in each interval, giving $r_{j_1}, \ldots, r_{j_m}$. Since (g_k) converges at each r_{j_i}, choose K such that $|g_k(r_{j_i}) - g_l(r_{j_i})| < \varepsilon/3$ for all $k, l \geq K$ and all i.

For any $t \in [a,b]$, choose r_{j_i} with $|t - r_{j_i}| < \delta$. Then for $k, l \geq K$:

$$\begin{aligned} |g_k(t) - g_l(t)| &\leq |g_k(t) - g_k(r_{j_i})| + |g_k(r_{j_i}) - g_l(r_{j_i})| + |g_l(r_{j_i}) - g_l(t)| \\ &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon. \end{aligned}$$

So $\|g_k - g_l\|_\infty < \varepsilon$ for $k, l \geq K$, showing (g_k) is Cauchy in $(C[a,b], \|\cdot\|_\infty)$, hence convergent.

($\Rightarrow$) **Relatively compact implies uniformly bounded and equicontinuous.**

Uniformly bounded: If $\text{cl}(\mathcal{F})$ is compact, it is bounded, so $\sup_{f \in \mathcal{F}} \|f\|_\infty \leq \sup_{g \in \text{cl}(\mathcal{F})} \|g\|_\infty < \infty$.

Equicontinuous: Given $\varepsilon > 0$, cover $\mathrm{cl}(\mathcal{F})$ by finitely many balls $B(g_i, \varepsilon/3)$, $i = 1, \ldots, N$. Each $g_i \in C[a,b]$ is uniformly continuous on $[a,b]$, so choose $\delta_i > 0$ with $|g_i(s) - g_i(t)| < \varepsilon/3$ when $|s-t| < \delta_i$. Set $\delta = \min(\delta_1, \ldots, \delta_N) > 0$. For any $f \in \mathcal{F}$, choose g_i with $\|f - g_i\|_\infty < \varepsilon/3$. For $|s-t| < \delta$: $|f(s) - f(t)| \le |f(s) - g_i(s)| + |g_i(s) - g_i(t)| + |g_i(t) - f(t)| < \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon$. □

Arzelà–Ascoli in Policy Optimization

In continuous-state RL, when we optimize over a class of value functions or policies, we need compactness to guarantee that maximizing sequences have convergent subsequences. The Arzelà–Ascoli theorem provides the criterion: if value function approximations are uniformly bounded and equicontinuous (e.g., Lipschitz with a common Lipschitz constant), then any sequence has a uniformly convergent subsequence. This is used in the analysis of fitted value iteration and actor-critic methods on continuous state spaces.

6.4 Compact Operators

Definition 6.5 (Compact Operator)**.** A linear operator $T\colon X \to Y$ between normed spaces is *compact* if T maps bounded sets to relatively compact sets, i.e., for every bounded $B \subseteq X$, the closure $\mathrm{cl}(T(B))$ is compact in Y. Equivalently, T is compact if every bounded sequence (x_n) in X has a subsequence (x_{n_k}) such that (Tx_{n_k}) converges in Y.

Proposition 6.7 (Compact Operators are Bounded)**.** *Every compact operator is bounded.*

Proof. Suppose T is compact but not bounded. Then there exist (x_n) with $\|x_n\| = 1$ and $\|Tx_n\| \to \infty$. Since (x_n) is bounded, compactness gives a subsequence with (Tx_{n_k}) convergent, hence bounded. But $\|Tx_{n_k}\| \to \infty$, a contradiction. □

Example 6.1 (Integral Operators are Often Compact)**.** The integral operator from Example **??** with continuous kernel k on $[a,b] \times [a,b]$ is compact as an operator $C[a,b] \to C[a,b]$. This follows from Arzelà–Ascoli: if (f_n) is bounded in $C[a,b]$, then (Tf_n) is uniformly bounded (since T is bounded) and equicontinuous (since k is uniformly continuous on the compact set $[a,b]^2$, and $|(Tf_n)(s_1) - (Tf_n)(s_2)| \le \|f_n\|_\infty \int_a^b |k(s_1,t) - k(s_2,t)|\,dt$ is small uniformly in n when $|s_1 - s_2|$ is small).

7 The Baire Category Theorem

The Baire category theorem is one of the most powerful existence results in analysis. It underpins the three fundamental theorems of Banach space theory: the Uniform Boundedness Principle, the Open Mapping Theorem, and the Closed Graph Theorem (all treated in the next lesson).

7.1 Nowhere Dense Sets and Category

Definition 7.1 (Nowhere Dense)**.** A subset A of a metric space X is *nowhere dense* if $\mathrm{int}(\mathrm{cl}(A)) = \emptyset$, i.e., the closure of A has empty interior. Equivalently, A is nowhere dense if and only if for every non-empty open set U, there is a non-empty open set $V \subseteq U$ with $V \cap A = \emptyset$.

Example 7.1. In $\mathbb{R}$: $\mathbb{Z}$ is nowhere dense, $\{1/n : n \in \mathbb{N}\}$ is nowhere dense, $\mathbb{Q}$ is *not* nowhere dense (its closure is $\mathbb{R}$, which has non-empty interior). The Cantor set is a closed, uncountable, nowhere dense subset of $[0,1]$.

Definition 7.2 (First and Second Category)**.** A subset of a metric space is of *first category* (or *meager*) if it is a countable union of nowhere dense sets. A set is of *second category* (or *non-meager*) if it is not of first category.

Remark 7.1. "First category" means "small" in the topological sense, and "second category" means "large." This is a different notion of smallness from measure zero.

7.2 Statement and Proof

Theorem 7.1 (Baire Category Theorem)**.** *Every complete metric space is of second category in itself. Equivalently, if (X, d) is a complete metric space and $X = \bigcup_{n=1}^{\infty} F_n$ with each F_n closed, then at least one F_n has non-empty interior.*

Proof. We prove the equivalent contrapositive: if $A_1, A_2, \ldots$ are nowhere dense subsets of a complete metric space X, then $\bigcup_{n=1}^{\infty} A_n \neq X$.

Since A_1 is nowhere dense, $X \setminus \mathrm{cl}(A_1)$ is a non-empty open set (if $\mathrm{cl}(A_1) = X$, then $\mathrm{int}(\mathrm{cl}(A_1)) = X \neq \emptyset$, contradicting nowhere density). Choose $x_1 \in X \setminus \mathrm{cl}(A_1)$ and $r_1 > 0$ with $\overline{B}(x_1, r_1) \subseteq X \setminus \mathrm{cl}(A_1)$ and $r_1 < 1$. In particular, $\overline{B}(x_1, r_1) \cap A_1 = \emptyset$.

Since A_2 is nowhere dense and $B(x_1, r_1)$ is a non-empty open set, there exists a non-empty open ball $B(x_2, r_2) \subseteq B(x_1, r_1)$ with $\overline{B}(x_2, r_2) \cap \mathrm{cl}(A_2) = \emptyset$ and $r_2 < 1/2$.

Continuing inductively, at step n we find x_n and $0 < r_n < 1/n$ with

$$\overline{B}(x_n, r_n) \subseteq B(x_{n-1}, r_{n-1}) \quad \text{and} \quad \overline{B}(x_n, r_n) \cap A_n = \emptyset.$$

The sequence (x_n) is Cauchy: for $m > n$, $x_m \in B(x_n, r_n)$, so $d(x_m, x_n) < r_n < 1/n \to 0$. Since X is complete, $x_n \to x$ for some $x \in X$.

For each n, $x_m \in \overline{B}(x_n, r_n)$ for all $m \geq n$ (by nesting), so $x = \lim x_m \in \overline{B}(x_n, r_n)$ (closed balls are closed). Therefore $x \notin A_n$ for any n, so $x \notin \bigcup_{n=1}^{\infty} A_n$. Hence $\bigcup_{n=1}^{\infty} A_n \neq X$. □

Corollary 7.2. *In a complete metric space, the intersection of countably many dense open sets is dense (in fact, it is a dense G_δ set).*

Proof. Let $G_1, G_2, \ldots$ be dense open sets in a complete metric space X. Their complements $F_n = X \setminus G_n$ are closed and nowhere dense (since G_n is dense, $\mathrm{cl}(F_n) = F_n$ has empty interior: any ball $B(x, r)$ contains a point of G_n, hence a point outside F_n).

By the Baire category theorem, $\bigcup F_n \neq X$, so $\bigcap G_n = X \setminus \bigcup F_n \neq \emptyset$.

To show density: let U be any non-empty open set. Then U is a complete metric space (as an open subset of X... actually we need a slight modification). Instead, apply the construction in the proof of Theorem 7.1 starting from any non-empty open ball $B(x_0, r_0) \subseteq U$: since each G_n is dense, $B(x_0, r_0) \cap G_n$ is a non-empty open set. Choose nested closed balls $\overline{B}(x_n, r_n) \subseteq B(x_{n-1}, r_{n-1}) \cap G_n$ with $r_n \to 0$. The limit $x = \lim x_n$ lies in $\overline{B}(x_0, r_0) \subseteq U$ and in each G_n, so $x \in U \cap \bigcap G_n$. □

Corollary 7.3 ($\mathbb{Q}$ is Not a G_δ in $\mathbb{R}$)**.** *The rationals $\mathbb{Q}$ cannot be written as a countable intersection of open sets.*

Proof. Suppose $\mathbb{Q} = \bigcap_{n=1}^{\infty} G_n$ with each G_n open. Since $\mathbb{Q}$ is dense, each G_n is dense. Also, for each $q \in \mathbb{Q}$, $\mathbb{R} \setminus \{q\}$ is open and dense. By Corollary 7.2, $\bigcap_n G_n \cap \bigcap_{q \in \mathbb{Q}} (\mathbb{R} \setminus \{q\}) = \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \emptyset$ is dense, which is a contradiction (the empty set is not dense in $\mathbb{R}$). □

7.3 Application Preview

Remark 7.2 (Preview: The Uniform Boundedness Principle). The Baire category theorem has the following powerful consequence (proved in the next lesson): if $\{T_\alpha\}_{\alpha \in I}$ is a family of bounded linear operators from a Banach space X to a normed space Y such that $\sup_\alpha \|T_\alpha x\| < \infty$ for every $x \in X$ (pointwise boundedness), then $\sup_\alpha \|T_\alpha\| < \infty$ (uniform boundedness). The proof works by writing $X = \bigcup_{n=1}^{\infty} \{x : \sup_\alpha \|T_\alpha x\| \leq n\}$ and applying Baire to conclude that one of these closed sets has non-empty interior.

Baire Category and Genericity in RL

The Baire category theorem is used in RL theory to prove "genericity" results: that certain properties hold for "most" value functions or policies in a topological sense. For instance, one can show that for a generic MDP, the optimal policy is unique (the set of MDPs with multiple optimal policies is meager). The Uniform Boundedness Principle, which follows from Baire, is used in the analysis of temporal-difference learning to conclude that pointwise convergence of TD estimates implies uniform bounds on approximation error.

8 Exercises

Exercise 8.1 (Metrics from Bounded Metrics)**.** Let (X, d) be a metric space. Show that $\tilde{d}(x, y) = \min(d(x, y), 1)$ and $\hat{d}(x, y) = d(x, y)/(1 + d(x, y))$ are both metrics on X that generate the same topology as d. Show that both $\tilde{d}$ and $\hat{d}$ are bounded metrics, and that (X, d) is complete if and only if $(X, \hat{d})$ is complete.

Exercise 8.2 (Non-Completeness of $C[0, 1]$). *Under*L^1] Give a detailed proof that $(C[0, 1], \|\cdot\|_1)$ is not complete by constructing an explicit Cauchy sequence that does not converge. (You may modify the construction in Example **??**, but provide full details.)

Exercise 8.3 (Closed Subspace of a Banach Space)**.** Let X be a Banach space and $Y \subseteq X$ a subspace (i.e., a vector subspace). Show that Y is a Banach space (under the restricted norm) if and only if Y is a closed subset of X.

Exercise 8.4 (Operator Norm Computation)**.** Let $T\colon (\mathbb{R}^2, \|\cdot\|_1) \to (\mathbb{R}^2, \|\cdot\|_\infty)$ be defined by $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2)$. Compute $\|T\|$ exactly.
Hint: Use the definition $\|T\| = \sup_{\|x\|_1 \le 1} \|Tx\|_\infty$ and parametrize the unit ball in $\|\cdot\|_1$.

Exercise 8.5 (Completeness of ℓ^∞)**.** Prove directly (without appealing to Theorem **??**) that $(\ell^\infty, \|\cdot\|_\infty)$ is a Banach space.

Exercise 8.6 (The Bellman Operator Norm)**.** Let $\mathcal{S} = \{1, \dots, n\}$ be a finite state space and $P \in \mathbb{R}^{n\times n}$ a stochastic matrix (non-negative entries, rows sum to 1). Define the linear operator $L\colon (\mathbb{R}^n, \|\cdot\|_\infty) \to (\mathbb{R}^n, \|\cdot\|_\infty)$ by $(LV)(s) = \gamma \sum_{s'} P(s'|s)V(s')$ for $\gamma \in (0, 1)$.

(a) Show that $\|L\| = \gamma$.
(b) Conclude that $T = r + L$ (the Bellman operator for a fixed policy) is a contraction with factor γ under $\|\cdot\|_\infty$.
(c) Verify the contraction property directly by computing $\|TV - TW\|_\infty$ for arbitrary $V, W \in \mathbb{R}^n$.

Exercise 8.7 (Riesz's Lemma is Sharp)**.** Show that in ℓ^2, Riesz's lemma can be strengthened to $\theta = 1$: for any proper closed subspace Y, there exists x with $\|x\| = 1$ and $d(x, Y) = 1$.
Hint: Use the orthogonal projection onto Y.

Exercise 8.8 (Equicontinuity and Compactness)**.** Let $K > 0$ and $\mathcal{F}_K = \{f \in C[0, 1] : |f(s) - f(t)| \le K|s - t| \text{ for all } s, t, \text{ and } |f(0)| \le 1\}$.

(a) Show that $\mathcal{F}_K$ is equicontinuous and uniformly bounded.
(b) Conclude by Arzelà–Ascoli that $\mathcal{F}_K$ is relatively compact in $(C[0, 1], \|\cdot\|_\infty)$.
(c) Show that $\mathcal{F}_K$ is in fact closed, hence compact.

Exercise 8.9 (Baire Category Application)**.** Use the Baire category theorem to show that $\mathbb{R}$ cannot be written as a countable union of sets, each of which is closed and has empty interior. Give an example showing that $\mathbb{R}$ *can* be written as a countable union of closed sets (some of which necessarily have non-empty interior).

Exercise 8.10 (Nowhere Differentiable Functions are Generic)**.** This exercise outlines the classical Baire category proof that "most" continuous functions are nowhere differentiable.

Let $A_n = \{f \in C[0,1] : \text{there exists } x \in [0,1] \text{ with } |f(x+h)-f(x)| \leq n|h| \text{ for all sufficiently small } |h|\}$.

(a) Show that A_n is a closed subset of $(C[0,1], \|\cdot\|_\infty)$.

(b) Show that A_n has empty interior. *Hint:* Approximate any $f \in A_n$ by a piecewise-linear function with slopes exceeding n in magnitude.

(c) Conclude that $\bigcup_{n=1}^{\infty} A_n$ is meager, so the set of functions that are somewhere differentiable is meager. The "typical" continuous function is nowhere differentiable.

9 Summary of Key Results

Result	**Statement (Informal)**	**RL/ML Relevance**
Metric space axioms (Def. **??**)	Distance function with positivity, symmetry, triangle inequality	Foundation for convergence analysis
Completeness of $\mathbb{R}^n$ (Thm. **??**)	Every Cauchy sequence in $\mathbb{R}^n$ converges	Finite-state value iteration converges
Completion theorem (Thm. **??**)	Every metric space embeds densely in a complete space	Justifies function approximation in incomplete spaces
Norm equivalence in $\mathbb{R}^n$ (Thm. **??**)	All norms on $\mathbb{R}^n$ are equivalent	Convergence in finite-state RL is norm-independent
Banach = complete normed (Def. **??**)	Banach spaces are the natural setting for fixed-point theorems	Value function spaces must be Banach for contraction arguments
Series characterization (Thm. **??**)	Absolute convergence $\Rightarrow$ convergence iff Banach	Practical completeness test
$C[a,b]$ is Banach under $\|\cdot\|_\infty$ (Thm. **??**)	Uniform limit of continuous functions is continuous	Value functions on compact state spaces form a Banach space
Bounded $\Leftrightarrow$ continuous (Thm. **??**)	A linear operator is bounded iff continuous	Bellman operator continuity
$B(X,Y)$ Banach when Y Banach (Thm. **??**)	Operator space inherits completeness	Space of operators on value functions is complete
Dual is always Banach (Cor. **??**)	X^* is Banach even if X is not	Policy gradient as element of dual space
Compactness equivalence (Thm. 6.2)	Compact $\Leftrightarrow$ seq. compact $\Leftrightarrow$ complete + totally bounded	Existence of optimal policies on compact spaces
Riesz's theorem (Thm. 6.5)	Closed unit ball compact $\Leftrightarrow$ finite-dimensional	Explains challenges in infinite-dim. function approximation
Arzelà–Ascoli (Thm. 6.6)	Equicontinuous + bounded $\Leftrightarrow$ relatively compact	Convergence of Lipschitz value function sequences
Baire category (Thm. 7.1)	Complete metric space $\neq$ countable union of nowhere dense sets	Underpins Uniform Boundedness Principle

Looking Ahead

With metric spaces, norms, completeness, operators, and compactness in hand, we are ready for the next lesson, which covers:

- The **Banach Fixed-Point Theorem** (contraction mapping theorem) — the core tool for proving convergence of value iteration, policy iteration, and TD-learning.
- The **Uniform Boundedness Principle** — used to establish uniform error bounds in function approximation.
- The **Open Mapping Theorem** and **Closed Graph Theorem** — structural results for bounded operators between Banach spaces.
- **Hilbert spaces** — inner product spaces where the Projection Theorem enables least-squares value function approximation.